Simulating Nonresidential Water Demand with a Stochastic End-Use Model

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Abstract: The end-use model SIMDEUM for residential water demand has been extended to incorporate nonresidential water demand. The model was developed to predict water-demand patterns with a small timescale (1 s) and small spatial scale (at the water meter connection). The end-use model is based on statistical information about users and end uses: data on occupancy; the frequency of use; duration and flow per water-use event; and the occurrence over the day of different end uses, such as flushing the toilet, doing the laundry, and washing hands. The model follows a modular approach, in that each type of building is composed of functional rooms, such as lodgings, restaurants, and conference rooms. A functional room is characterized by its typical users and water-using appliances. With this approach, nonresidential buildings' water-demand patterns over the day can be simulated. The simulation results for an office building, a hotel, and a nursing home were compared to measured water-demand patterns with regard to attributes such as peak flow and daily total water use, as well as the shape of the pattern. The simulation results show a good correspondence to measured water demands. The end-use model is based on independent statistical information and not on flow measurements. The input parameters are available before any information on annual or daily water use is available; the parameters are not fitted on flow measurements. Therefore, the model is transferable to a diverse range of nonresidential water-demand types. The model can be applied in the design stage (prebuild), in scenario studies, and in distribution network models. **DOI: 10.1061/(ASCE)WR.1943-5452.0000146.** © *2011 American Society of Civil Engineers.*

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Introduction

A water-quality network model of drinking-water distribution systems requires a hydraulic model with an accurate demand allocation on small spatial (per connection) and temporal (a per second or per minute basis) scales (Blokker et al. 2008b). Furthermore, there is an interest in the Netherlands in designing self-cleaning water-distribution networks. This requires that during maximum demand, a certain threshold velocity is exceeded (Vreeburg 2010). For the design of a self-cleaning network, it is important to predict the maximum water demand as well as the daily demands. With the stochastic end-use model SIMDEUM (Blokker et al. 2010c), it is possible to generate high-resolution water-demand patterns for residential users and use them in water-quality modeling (Blokker et al. 2010a, 2011) to design self-cleaning networks (Blokker 2010; Blokker et al. 2010b). For nonresidential users, no high-resolution water-demand model is available.

In the Netherlands, 72% of total water demand is residential, 12% is for small-scale business users, and 16% is for large-scale business users (Geudens 2008). Small-scale business users are defined as nonresidential customers who use less than 10,000 m^3/year. These users include cattle farms, shops, offices, campgrounds, holiday resorts, small industries, schools, health care institutions, and offices (Baggelaar and Geudens 2008). Large-scale business users are defined as customers who use more than 10,000 m^3/year. The water demand of large-scale business users is measured by the majority of Dutch water companies on relatively small timescales (5 min). These water-demand patterns are user-specific and may be relatively constant; therefore, a stochastic water-demand model would not be favorable. Because measurements are available, the need for a water-demand model is not very large. For small-scale businesses, there is a need for a high-resolution, nonresidential water-demand model. According to Dutch drinking-water companies and building installation sector offices, hotels and nursing homes are most prevalent (Pieterse-Quirijns et al. 2009).

For temporal high-resolution residential water demand, two types of demand models are available, namely, the Poisson rectangular pulse (PRP) model (Buchberger et al. 2003) and the end-use model SIMDEUM (Blokker et al. 2010c). The PRP model requires a measurement campaign to establish the values of the PRP parameters. The PRP model works well as a descriptive model; however, it does not provide insight into the factors that influence the water demand (Blokker et al. 2008a). The end-use model SIMDEUM is a physical model that requires information on water-using appliances and users. The input parameters include the water use per flush for several types of toilets, the average and variance of the duration of taking a shower, the household composition, and statistics that describe the sleep-wake rhythm of each individual user. Through Monte Carlo simulations, it provides a water-demand pattern of a

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single house at a timescale of 1 s. Blokker et al. (2010c) showed that the end-use model describes and predicts residential drinking water demand very well.

Because the small-scale business users are diverse in type and size, the PRP model would require measurements for all these types and sizes. The end-use model, however, offers a relatively easy way to construct a water-demand model. In this paper, an end-use model for nonresidential water demand is developed and discussed. The end-use model SIMDEUM for residential water use was expanded to include nonresidential water demand for new types of appliances, such as coffee machines, and new types of users, such as hotel guests and office employees. Compared to the existing model, specific nonresidential water-demand parameters were introduced, and a new approach toward determining the time of water use over the day was developed. By introducing the concept of functional rooms, which are characterized by their water-using appliances and specific type of users, it is possible to model various types of nonresidential water demands. The model was tested for office buildings, hotels, and nursing homes. The results are compared to measurement data.

Methods and Materials: Model Development

Basic Model

The end-use model produces a water-demand pattern that can be described by the following equations (Blokker et al. 2010c):

$$Q = \sum_{k=1}^{M} \sum_{j=1}^{N} \sum_{i=1}^{F_{jk}} B(I_{ijk}, D_{ijk}, \tau_{ijk}) \quad (1)$$

$$B(I_{ijk}, D_{ijk}, \tau_{ijk}) = \begin{cases} I_{ijk} & \tau_{ijk} < T < \tau_{ijk} + D_{ijk} \\ 0 & \text{elsewhere} \end{cases} \quad (2)$$

Here, i, j, and k are indices; k counts all end uses from 1 to M; j counts all users from 1 to N; and i counts all water-use occurrences per end use from 1 to F_{jk} (the frequency of use for user j and end use k). D = pulse duration (in s); I = pulse intensity (flow in L/s); and τ = time at which the tap is opened. Thus, D_{ijk} = duration for end-use type k for user j and occurrence i. $B(I, D, \tau)$ = a block function, which equals I at time τ to $\tau + D$ and 0 during the rest of the day. The summation is done for all (M) available end uses, all (N) users, and per end use for the frequency of use (F); this leads to the total water-demand pattern Q (L/s) over the day.

In the case of nonresidential water demand, Q in Eq. (1) is determined per functional room in the building and then summed

$$Q = \sum_{h=1}^{L} Q_h \quad (3)$$

Here h is an index that counts the functional rooms 1 to L.

All parameters are described by probability-distribution functions (PDFs), which are different for each end use.

Functional Rooms (h)

For office buildings, hotels, and nursing homes, the following functional rooms were identified: meeting area, lodging, restaurant, fitness room, and other (e.g., technical rooms). Each functional room has its own end-use types and users (Tables 1 and 2). A meeting area contains toilets and coffee machines, and the users are employees and visitors. A lodging is a hotel room or a room in a nursing home, including bathrooms. These functional rooms involve a water closet (WC), a hot and cold water tap, a shower and/or bath, and users with water-related behavior that is more or less residential. The difference between lodging in hotels and nursing homes is that in hotels, the water use is determined by the hotel guests; in a nursing home, the water use is determined much more by the caretakers than by the residents themselves. A restaurant can be a cafeteria in an office building or nursing home or a full-service restaurant in a hotel, and is defined by a kitchen tap and dishwashers. In the kitchen, the users are the kitchen personnel. A fitness room in a hotel or office building contains a shower and potentially the same users as the meeting room, but the frequency of use will be less than for residential showers. The "other" category includes industrial washing machines, hot water taps to fill buckets for cleaning, and other services. Its users use water as part of their jobs with demand for laundry and cleaning activities and the associated water use is driven by other users and activities in building, namely, guests, customers, and so forth.

End Uses (k)

On the basis of the residential water-demand model, eight main types of end uses are defined at the fixture or appliance level: WC, shower, washing machine, dishwasher, kitchen tap, bathroom tap, bathtub and outside tap. For a model of nonresidential water demand, these types are extended to include a coffee machine. Outdoor water use was not considered, because in the Netherlands its contribution in nonresidential buildings is limited. The number of installed appliances in a functional room is also of importance. This is denoted by the penetration rate, i.e., the number of appliances of the different types: for example, six WCs and six cold water taps in the meeting-area functional room. Nonresidential end uses are described by fixed input parameters rather than statistical distributions.

Each type of end use contains different subtypes that constitute the end use. For a WC, this can be an old-fashioned toilet with a large cistern consuming 9–12 liters, or a new dual-flush toilet consuming only 6 liters with a water-saving possibility of consuming only 3 liters. For the tap, the subtype is defined by its uses, such as washing hands and brushing teeth. For a model of nonresidential water demand, other subtypes other than the residential water-demand model appear, such as industrial dishwashers. The penetration rate of the subtypes in a functional room is the fraction of

Table 1. Types of Users (and Reason for Variable Occupancy) Who Take Water in Functional Rooms

Functional room	Building type		
	Office	Hotel	Nursing home
Meeting area	Employees (full-time/part-time working), visitors	Guests (meeting, conference, theater), employees	Visitors, employees
Lodging	—	Hotel guests (tourists, business people)	Residents
Restaurant	Kitchen personnel	Kitchen personnel	Kitchen personnel
Fitness room	Employees using the fitness room	Hotel guests using the fitness room	—
Technical/other	Number of buckets filled for cleaning, washing machines, and so forth		

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Table 2. Frequency, Duration and Intensity for Several Types and Subtypes of End Uses in Different Functional Rooms: Average (μ) and Probability Distribution Function (PDF)

Functional room	End-use type/subtype		Frequency (person^{-1}·day^{-1})[a]	Duration		Intensity (L/s)	
			μ	μ	PDF	μ	PDF
Meeting area	WC	6-liter cistern	4.0	2.4 min[b]	Not applicable	0.042	Not applicable
		9-liter cistern		3.6 min[b]			
		Urinal[c]		9 s		0.167	
	Tap	Washing hands	4.5	16 s	Not applicable	0.083	Not applicable
	Coffee machine[d]		8.0	4.8 s	Not applicable	0.042	Not applicable
Lodging	Bathtub	120 liters	0.2	10 min	Not applicable	0.200	Not applicable
	WC	6-liter cistern	4.0	2.4 min[b]	Not applicable	0.042	Not applicable
		9-liter cistern		3.6 min[b]			
	Tap	Washing and shaving	6.0	40 s	Lognormal	0.083	Uniform
		Brushing teeth		15 s			
		Other		45 s			
	Shower	Normal	0.2–0.8[e]	8.5 min	χ^2	0.142	Not applicable
		Water-saving type				0.123	
		Comfort				0.365	
Restaurant	Dishwasher	Brand and type	Variable[f]	Specific dishwashing pattern (2–14 L per cycle, 50 s)			
	Kitchen tap	Cold/short	Variable[g]	15 s	Lognormal	0.167	Uniform
		Cold/long		60 s		0.250	
		Hot/short		15 s		0.167	
		Hot/long		60 s		0.25	
Fitness room	Shower	Normal	0.8	8.5 min	χ^2	0.142	Not applicable
		Water-saving type				0.123	
Technical/other	Cleaning	Filling bucket	Variable[g]	30 s	Not applicable	0.500	Not applicable
	Washing machine	Brand and type	Variable	Specific washing pattern (9–200 L per cycle)			

[a]Frequency is Poisson distributed.
[b]With a water-saving option, the duration is reduced to 50% of the original value for 80% of flushes.
[c]Frequency of use for men only; toilet/urinal division is 1/3.
[d]Also tea, water, and so forth. Water use is the filling of the reservoir.
[e]0.2 at nursing homes; 0.8 in hotels.
[f]Frequency for dishwasher (per guest) depends on the cutlery and crockery used in restaurant and on the size of the dishwasher; frequency for kitchen tap depends on the number of dishes prepared per guest.
[g]Frequency for cleaning is the number of buckets filled; this relates to floor space.

appliances that constitute the type (e.g., type of toilet) or the use per type (e.g., hand-washing use for the tap in lodging functional rooms).

Users (j)

Users are key in an end-use model because they induce the end uses. Users are defined per functional room. Unlike the residential end-use model, users are not defined as individuals, but as a group. In the model, properties of a nonresidential user group are its diurnal pattern (see related section) and the number of male and female individuals. A gender distinction was made because it influences toilet use; specifically, when a functional room has urinals and toilets. The diurnal pattern is related to the presence of users over the day. It represents the probability-distribution function of water-use events over the day for the group of users. Table 1 summarizes the functional rooms and the types of users for office buildings, hotels, and nursing homes.

The number of people present in the office on any one day will vary. In the Netherlands, 92% of men work full time (40 h per week); the remaining 8% work on average 17.4 h (standard deviation 8.1 h). Women on average work 25.5 h (standard deviation 10.9 h; Bosch et al. 2010). The working hours per week follow a normal probability distribution (Pieterse-Quirijns et al. 2009). In the simulation, the total occupancy per day is determined as follows:

- Of the male employees, 92% are present as a base value. For the remaining 8%, first the number of working hours per week is determined ($\mu = 17.4$; $\sigma = 8.1$). Next, for the subjects of whom the part-time factor (the number of working hours per week divided by 40) exceeds a random number r ($\mu = 0.6$; $\sigma = 0.1$) the presence for the simulated day is set to 1; for the others it is set to 0. Total presence is the sum of the base value and the number of part timers present.
- For the female employees, the number of working hours per week is determined ($\mu = 25.5$; $\sigma = 10.9$). Next, for the subjects for whom the part-time factor exceeds r, the presence for the simulated day is set to 1; for the others it is set to 0. Total presence is the sum of the number of part timers present.
- The number of visitors is determined from a normal probability distribution (μ = average number of visitors; $\sigma = 0.2^*\mu$) and divided by 2, because most visitors do not stay all day. The ratio between male and female visitors is the same as the ratio for employees.

The number of hotel guests occupying a hotel room during a simulation is determined in the following way:

- A room is occupied or not: the mean occupancy rate is determined from a beta probability distribution. The beta probability distribution is used because the beta random variable is by definition between 0 and 1; the parameters for the beta probability-density function can be determined from the mean and standard

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deviation. The mean occupancy is an input parameter and can be different for weekdays and the weekend; the standard deviation is assumed to be equal to one quarter of the mean value.

- An occupied room is used by either one person or two persons. In business hotels, approximately 60% of the occupied rooms are used by only one person during the week, whereas on the weekend and in tourist hotels, approximately 60% of the occupied rooms accommodate two people (Pieterse-Quirijns et al. 2009).
- The number of visitors is assumed to be a fixed percentage of the lodging guests.

The number of residents in a nursing home is assumed to be more or less stable. Given the fact that there are waiting lists for nursing homes, the occupancy is assumed to be 100%. For other types of users, such as kitchen personnel, there is even less information available on the variability of their presence. Therefore, in the water-demand model the number of users in the restaurant, fitness room, and technical room is assumed to be a fixed percentage of the number of residents.

Frequency of Use (f)

For each end use, the frequency of use was determined on the basis of either the information of the frequency for residential water demand or on expert opinion. The frequency is the number of uses per person per day. In the model, the kitchen tap, dishwasher, and washing machine are operated by the kitchen personnel, and the frequency is determined by other factors. The frequency of use for the dishwasher in a restaurant, for example, depends on the number of guests and the amount of cutlery and crockery that needs to be washed, which is different for each type of meal that is being served.

Table 3. Statistics for Residential Diurnal Patterns (Data from Pieterse-Quirijns et al. 2009)

Description		Tourist	Business traveler	Residents in retirement home
Time of getting up	μ	8:00	7:00	8:00
	σ	1:30	1:00	1:00
Time of leaving	μ	10:00	8:30	13:00
the house	σ	2:00	1:00	3:00
Duration of being	μ	14.0	9.0	10.0
away (h)	σ	1.0	2.0	4.5
Duration of sleep (h)	μ	8.0	8.0	8.0
	σ	1.0	0.5	1.0

The frequency of use is described by a discrete statistical distribution, i.e., the Poisson distribution. For nonresidential water demand, hardly any specific statistical information is available for the frequency of use. The Poisson distribution is assumed for all end uses because it was applicable for the frequency of most residential water-demand end uses. Table 2 summarizes for all end uses how f is described.

Pulse Intensity (I) and Pulse Duration (D)

For each end use, the flow (in L/s) and duration (in s) is based on information of residential end uses and on technical information on water-using appliances collected from installation guides and an Internet search (Pieterse-Quirijns et al. 2009). The duration of taking a shower is typically determined by the user, whereas the duration of filling the toilet cistern is determined by the volume of the cistern and the inlet flow, which is related to the indoor plumbing. The duration of taking water from a bathroom or kitchen tap, which is determined by the user, can be described using a lognormal distribution function determined from a statistical analysis of survey data from 3,200 respondents (Blokker et al. 2010c). The same statistical distributions as the residential water-demand end uses are assumed in this model. Table 2 summarizes for all end uses how I and D are described.

Diurnal Pattern, Time of Water Use (τ)

There is no survey available that addresses the time of the day when a water-use event is likely to take place for all end uses. For the residential end users, the time of day of water use is strongly related to whether people are at home and whether they are asleep, getting up, or preparing for bed (Blokker et al. 2010c). This relationship is also used for the lodging functional rooms in a hotel. For hotel guests, a sleep-wake rhythm is applied (Table 3) that differs from residential users' diurnal pattern. Different diurnal patterns are applied for tourists and business travelers. For nonresidential users, such as employees or restaurant guests, and for nursing home residents, the diurnal patterns are related to presence during (for example) lunch hours, defined by $t1 - t4$ (Table 4). Time $t1$ and $t4$ denote the edges of people's presence; between $t2$ and $t3$ a greater probability of use occurs. These parameters were derived from specific surveys (Pieterse-Quirijns et al. 2009); see also the Results section.

A diurnal pattern for a residential-like user is constructed as described by Blokker et al. (2010c). The diurnal pattern is the basis of the probability-density function of water-use events. Because the times of waking up, leaving the home, returning home, and going to bed vary (Table 3), the probability of water-use events varies as well. A random time of water use (τ) can be retrieved from the normalized cumulative distribution function (CDF). An example

Table 4. Statistics for Nonresidential Diurnal Patterns (Data from Pieterse-Quirijns et al. 2009); Times $t1$–$t4$ Are Related to the Start and End of Water-Use Probability

		Office employee	Office cafeteria personnel	Office cleaner	Hotel conference guest	Hotel restaurant personnel	Hotel cleaner	Nursing home residents	Nursing home employees and visitors
$t1$	μ	8:00	8:00	17:45	8:00	7:30	9:00	6:30	6:00
	σ	1:00	1:00	0:05	1:00	1:00	1:00	1:00	2:00
$t2$	μ	12:15	12:30	—	11:00	10:00	—	7:30	8:00
	σ	0:30	0:30	—	1:00	1:00	—	0:30	0:30
$t3$	μ	13:00	13:30	—	20:00	20:00	—	22:00	21:00
	σ	1:00	1:00	—	1:00	1:00	—	1:00	0:30
$t4$	μ	18:00	15:00	18:15	23:00	22:30	13:00	22:30	22:00
	σ	1:00	0:15	0:05	1:00	0:30	1:00	1:00	2:00

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is shown in Fig. 1(a), in which a random number equal to 0.45 indicates that τ for the water-use event is 18:15.

For the group of nonresidential users, the diurnal pattern is constructed in a similar manner, from the given μ and σ of $t1$ to $t4$:

- Around $t1$ and $t2$, following a normal distribution, a PDF is constructed that equals 0 at $\mu - 3\sigma$ and is at its maximum at $\mu + 3\sigma$.
- Around $t3$ and $t4$, following a normal distribution, a PDF is constructed that is at its maximum at $\mu - 3\sigma$ and equals 0 at $\mu + 3\sigma$.

The CDF is the cumulative sum of the four PDFs and is scaled to be between zero and one. A random time of water use (τ) can be retrieved from the CDF. An example is shown in Fig. 1(b), in which a random number equal to 0.3 indicates that τ for the water-use event is 11:15.

Methods and Materials

Simulation

Once the statistical information on users and water outlets per functional room are put into the model, the simulation can be run. The output of the simulation is a time series of flows: a water-demand pattern. First, for each simulated functional room, the gender and occupancy (Table 1) and diurnal patterns of the occupants (Tables 3 and 4) are determined. Next, for each simulated functional room it is determined what end uses ($k = 1$ to M) and subtypes are present and how many. Then, per end use k (in random order), the frequency of use (F_{jk}) for all users ($j = 1$ to N) in the functional room is determined from the Poisson distribution function (Table 2). The frequency of use is divided over the total of number of appliances in a group (e.g., all ladies' WCs). After that, for all occurrences ($i = 1$ to F_{jk}), the duration and intensity are determined from the appropriate PDF (Table 2), and the time during the day (τ) that the water-use event occurs is determined from the diurnal PDF (Fig. 1). When the time τ_{ijk} is established for end use k, user j, and occurrence i, the time $\tau_{ijk} - D_{ijk} < T < \tau_{ijk} + D_{ijk}$ is blocked for end-use k, user $j + 1$ to N, and occurrence $i + 1$ to F_{jk}. This ensures that there is no overlapping use by multiple users of one end use, e.g., when only a single shower is present, the use of it by one person cannot coincide with its use by a second person. However, user j could potentially draw water from multiple end uses (k and $k + 1$) at the same time. This is not actively blocked, but seldom seems to occur in residential water use, and is of less importance with the nonresidential users. The sum of all water demands is the water-demand pattern of the functional room; the sum of the water demands of all functional rooms is the water-demand pattern of the nonresidential building.

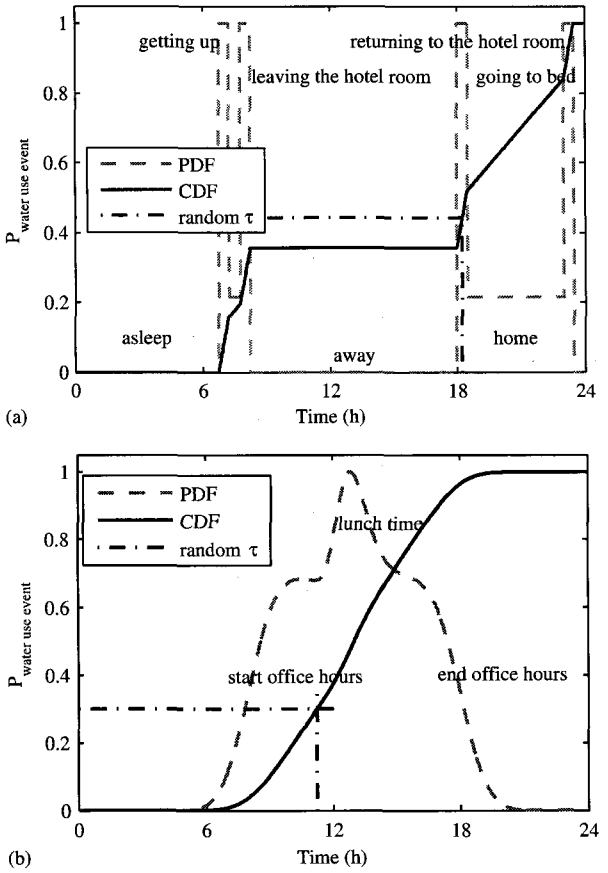


Fig. 1. An example of drawing a random time (τ) from the normalized CDF of water use for (a) a business traveler with a residential diurnal pattern and (b) an office employee with a nonresidential diurnal pattern

The output of one simulation run is one water-demand pattern. By performing more simulations, multiple patterns (days) of a specific building are found. Owing to the stochastic nature of water-demand patterns, this single simulated water-demand pattern is only one possible outcome. With a Monte Carlo simulation, many simulation results are obtained, and a statistical analysis can be made on the results. The end-use model was programmed in MATLAB to run the simulations.

Model Validation

To validate the water-demand model, the results are compared to water-demand measurements on several characteristic parameters. Because the output of the model is stochastic in nature, the cumulative frequency distribution of the characteristic parameters is used in the comparison. To quantify the validation, some statistic measures are used.

For the residential water-demand model, eight different characteristic parameters were used that are suitable for different timescales and spatial scales (Blokker et al. 2010c). For the validation of the nonresidential water-demand model of an office building, hotel, and nursing home, the characteristics Q_{max} (the maximum flow for different timescales), V (the total volume per day in m^3), Q_{day} (the demand pattern over the day in m^3/h), and C_d (the normalized multiplier pattern; Q_{day} divided by the average of Q_{day}) are used.

The parameters Q_{max} and V are presented as cumulative frequency distributions. The goodness of fit for the cumulative frequency distributions is determined with the mean error (ME), root-mean-square error (RMSE), and R^2 evaluated at each 10th integer percentile between 10 and 100 (10, 20, 30, ..., 100). Q_{day} is a pattern; the goodness of fit is expressed as the ME, RMSE, and R^2 of Q_{day}. The ME and RMSE are presented as absolute values and as percentages of the means of the measurements.

An analysis was done to determine which functional room contributes most to the total water use. The information on the most-contributing functional room has to be more reliable than the information about functional rooms that contribute only marginally to the total water use. Pieterse-Quirijns et al. (2009) did a more extensive analysis for three more office buildings, three more hotels, and two more nursing homes (without assisted living).

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Results

To establish the performance of the water-demand model, the results are compared to water-demand measurements. Demand measurements on a 5-min timescale were collected for office buildings, hotels, and nursing homes from several Dutch water companies. Specific input parameters for these buildings were gathered from surveys. These surveys inquired about the number of employees and visitors, the number of toilets, and so forth. In total, water-demand measurements and surveys of four office buildings and four hotels were collected; for the nursing homes, only one survey was returned, and measurements of three other nursing homes were supplied by a water company. As an illustration, the results of one office building, one hotel, and one nursing home are presented here. These are representative for the other buildings that were tested (Pieterse-Quirijns et al. 2009). For all three types of nonresidential buildings, the simulation approach and comparison figures and tables are kept as similar as possible.

Office Building

The particular office building that was simulated and validated is an office of a Dutch water company. Measurements were available on a 1-min time basis for 2 weeks (10 weekdays) in November 2008.

In this office, 284 men and 189 women are employed; on average 50 visitors per day are received. The office contains 38 toilets (7 urinals, 18 toilets for men, and 13 toilets for women), 4 showers, and 15 coffee machines. The kitchen is open for lunch, and 125 people make use of it. The kitchen contains one dishwasher (frequency is set to 0.3 per visitor for simple meals) and some kitchen taps for hot and cold water. The office is cleaned every day at the end of the day (after 17:00 hrs.) by 12 people filling 12 buckets of warm water and 12 buckets of cold water at the start of their shift.

For the simulation, 50 Monte Carlo simulation runs were done, and thus 50 weekday patterns were constructed for this office building. The maximum flow (Q_{max} per 5 min and per hour), the volume (V per day) and the pattern over the day (Q_{day}) for measured and SIMDEUM water-demand patterns are compared in Figs. 2 and 3 and Table 5.

Hotel

The simulated hotel also provides conference rooms and a theater. Measurements were available on a 5-min time basis for 30 days in June 2006.

The hotel has 112 rooms with an average occupancy of 80% during the week (60% one person, 40% two persons) and 40% during the weekend (40% one person, 60% two persons). The room contains a toilet, a tap for cold and hot water, a shower, and a bath.

For the conference rooms and theater, the hotel has 120 toilets (40 urinals, 40 toilets for men, and 40 toilets for women), 69 washstands and seven coffee machines. Conference visitors ($\mu = 350$, $\sigma = 100$) are present during the day; theater visitors ($\mu = 350$, $\sigma = 100$) are present during the evening.

The hotel serves breakfast, lunch, and dinner for on average 92, 150, and 100 guests, respectively. In the kitchen, water is used for food preparation and dishwashing. The kitchen contains one dishwasher; its frequency of use per guest is estimated at 0.2 for breakfast, 0.3 for lunch and 0.5 for dinner.

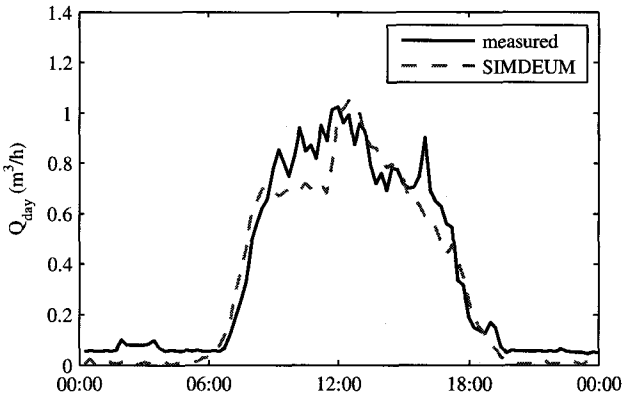


Fig. 3. Office building: Comparing average (Q_{day}), at a timescale of 15 min, of measurements (10 weekdays) and simulations (50 days)

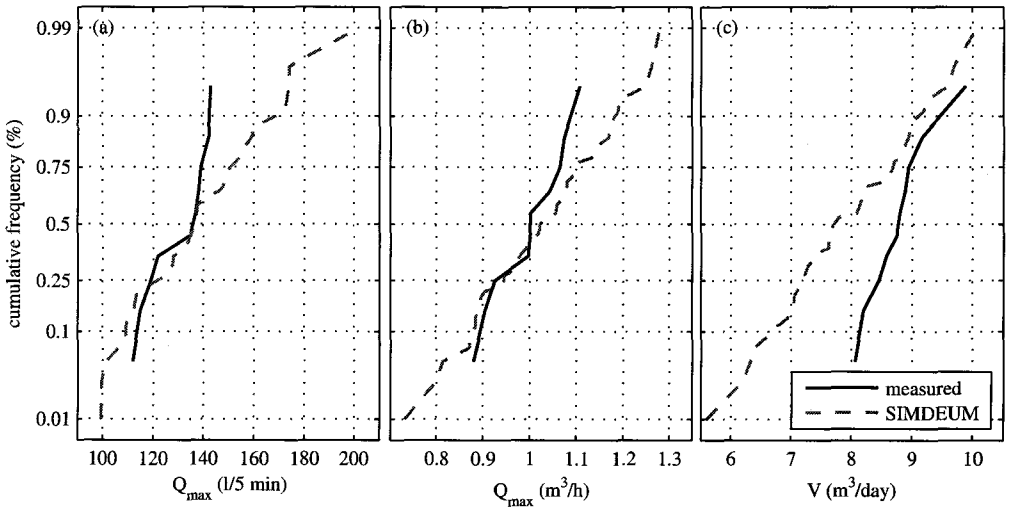


Fig. 2. Office building: Comparing measurements (10 weekdays at 1-min timescale) and simulations (50 days at 1-s timescale); (a) Q_{max} per 5 min, (b) Q_{max} per h, (c) V per day

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Table 5. Statistics of Comparison between Measured and Simulated Water Demand; All Values Are Statistically Significant ($\rho < 0.05$)

Building type	Figure	Q/V	Unit	ME	ME (%)	RMSE	RMSE (%)	R^2
Office	Fig. 2	Q_{max}	(l/5 min)	11.5	(8.7%)	22.4	(17.0%)	Not applicable
		Q_{max}	(m^3/h)	0.0	(4.0%)	0.1	(7.1%)	Not applicable
		V	(m^3/day)	−0.8	(−8.5%)	1.0	(10.9%)	Not applicable
	Fig. 3	Q_{day}	(m^3/h)	−0.0	(−10.1%)	0.1	(25.0%)	0.93
Hotel	Fig. 4	Q_{max}	(l/5 min)	−10.8	(−1.8%)	41.2	(6.9%)	0.91
		Q_{max}	(m^3/h)	−0.4	(−10.0%)	0.6	(12.6%)	0.62
		V	(m^3/day)	−3.6	(7.6%)	5.4	(11.3%)	0.63
	Fig. 5	Q_{day}	(m^3/h)	−0.1	(−7.4%)	0.5	(28.6%)	0.61
Nursing home	Fig. 6	Q_{max}	(l/5 min)	−29.6	(−7.2%)	38.1	(9.3%)	0.71
		Q_{max}	(m^3/h)	−0.7	(−19.2%)	1.0	(26.8%)	Not applicable
		V	(m^3/day)	−0.7	(2.3%)	2.9	(9.3%)	0.47
	Fig. 7	Q_{day}	(m^3/h)	0.1	(4.3%)	0.3	(22.9%)	0.88

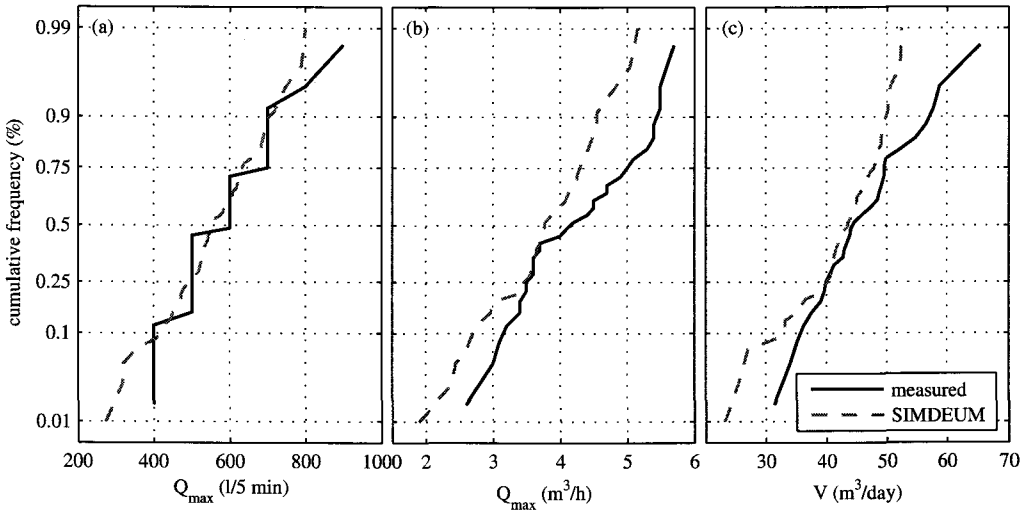


Fig. 4. Hotel: Comparing measurements (30 days at 5-min timescale) and simulations (50 days at 1-s timescale); (a) Q_{max} per 5 min, (b) Q_{max} per h, (c) V per day

The hotel cleaning is done in the morning by approximately 20 cleaners, taking hot and cold water for cleaning the hotel rooms and conference rooms.

For the simulation, 50 diurnal patterns were constructed for this hotel. The maximum flow (Q_{max} per 5 min and per hour), the volume (V per day) and the pattern over the day (Q_{day}) for measured and SIMDEUM water-demand patterns are compared in Figs. 4 and 5 and Table 5.

Nursing Home

The nursing home that was simulated has 110 single rooms in the nursing home itself and 80 apartments for assisted living. Measurements were available on a 5-min time basis for 30 days in June 2006. No survey was filled out; specific information was estimated from the nursing home's website and a survey of another nursing home.

The apartments for assisted living house one or two people over 65 years of age. They have a residential water-demand through toilet, shower, kitchen and bathroom tap, and washing machine use.

In the nursing home, a maximum of 350 people are present during the day. This is a combination of residents (110) and personnel and visitors (240). For personnel and visitors, 40 toilets and 40 washstands are available, as well as 10 coffee machines. The single rooms each have a toilet, a washstand, and a shower. Each floor (five in total) has a small kitchen to prepare meals and do washing up, which uses a small amount of water. The restaurant has a large

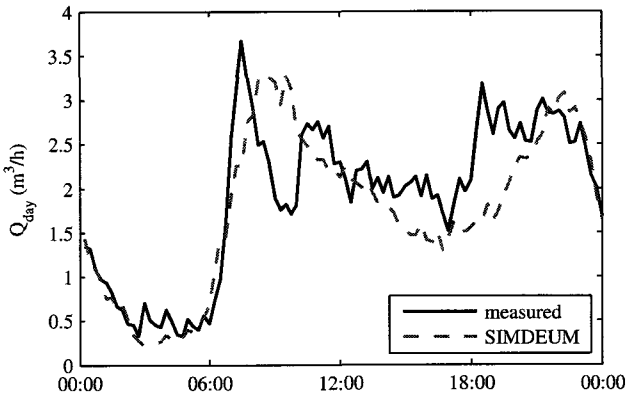


Fig. 5. Hotel: Comparing average Q_{day}, at a timescale of 15 min, of measurements (30 days) and simulations (50 days)

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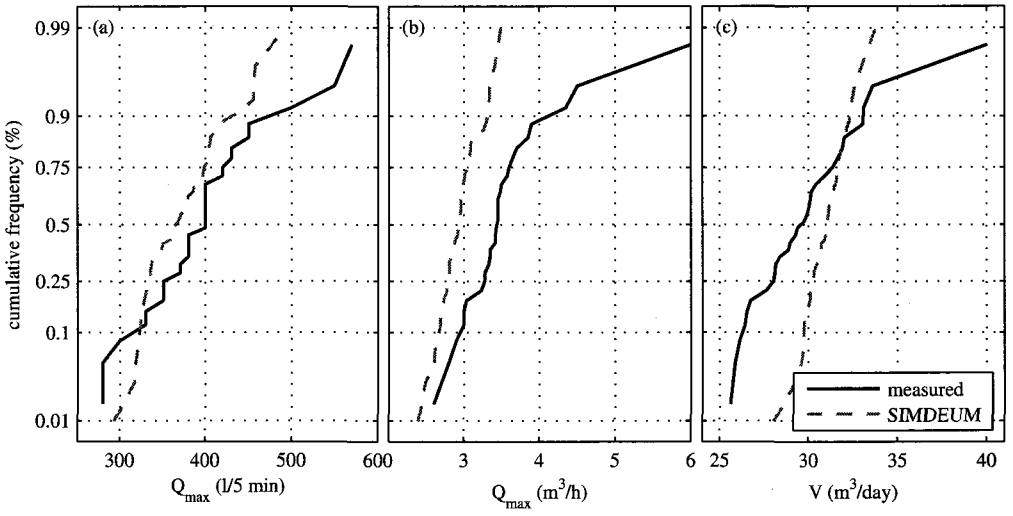


Fig. 6. Nursing home: Comparing measurements (30 days at 5-min timescale) and simulations (50 days at 1-s timescale); (a) Q_{max} per 5 min, (b) Q_{max} per h, (c) V per day

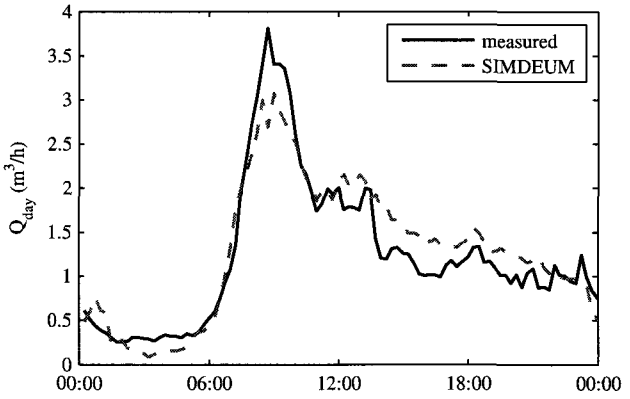


Fig. 7. Nursing home: Comparing average Q_{day}, at a timescale of 15 min, of measurements (30 days) and simulations (50 days)

dishwasher that runs approximately 120 times per day. Extra water use is induced by cleaning (approximately 20 cleaners during the entire working day), a hairdressing facility (approximately 40 water-use events per day), washing machines (approximately six runs per day), and bedpan cleaning.

For the simulation, 50 diurnal patterns were constructed for the nursing home plus assisted-living apartments. The maximum flow (Q_{max} per 5 min and per hour), the volume (V per day), and the pattern over the day (Q_{day}) for measured and SIMDEUM water-demand patterns are compared in Figs. 6 and 7 and Table 5.

Analysis of Contribution of Functional Rooms

Fig. 8 shows how much each functional room contributes to the total water use in an office, a hotel, and a nursing home.

Discussion

The simulations with the presented nonresidential SIMDEUM were compared with measured nonresidential water demand for the three presented buildings (Figs. 2–7; Table 5) and for three more office buildings, three more hotels and two more nursing homes (Pieterse-Quirijns et al. 2009). Figs. 2, 4, and 6 compare the maximum and total demand of simulations and measurements. Three things should be considered when interpreting these figures. First, the x-axes typically do not start at 0. Second, a normal probability plot emphasizes the extremities, i.e., the high and low percentiles. Third, for offices (Fig. 2), more simulation data than measurement data were plotted. The results are quantified in Table 5. Here, the comparison was done per 10th percentile; this indicates that the 1st and 99th percentiles were not taken into account.

For office buildings, the maximum flow and the average flow pattern over the day is typically predicted well by SIMDEUM (Pieterse-Quirijns et al. 2009), i.e., ME and RMSE $< 30\%$, and $R^2 > 0.7$. For hotels also, the maximum flow and the average flow pattern over the day is typically predicted well by SIMDEUM. This is the case for the hotel with conference rooms and a theater (which has relatively high water use in the evening) as well as for more common tourist hotels (Pieterse-Quirijns et al. 2009). For nursing homes, the maximum flow per 5 min and the average flow pattern over the day is typically predicted well by SIMDEUM. This is particularly remarkable, because only limited information of the nursing homes was available, because no surveys were filled in. The diurnal pattern for nursing homes without assisted living looks different from Fig. 7, in that the low night use starts earlier (before midnight); this is the case for both measured and simulated water demand (Pieterse-Quirijns et al. 2009).

The analysis per functional room (Fig. 8) showed that in office buildings, the meeting area is the main contributor; more specifically, the toilets in the meeting area. It is therefore important to have reliable information on toilet use: how many people are employed, the frequency of use, how much water is used per flush, whether a water-saving option is available, and how often this water-saving option is used. In hotels, the main contributor is the sum of hotel rooms. It is therefore important to have reliable information on the rooms: how many rooms are present, what the occupancy is, and what type of shower and toilet are installed (i.e., water-saving device or luxurious shower). For this particular hotel, the meeting area is also a large contributor, because it also serves as a conference hotel and hosts a theater. The water use in the presented nursing home is not determined by a single dominant functional room. The water use is largely determined by the toilet use of personnel

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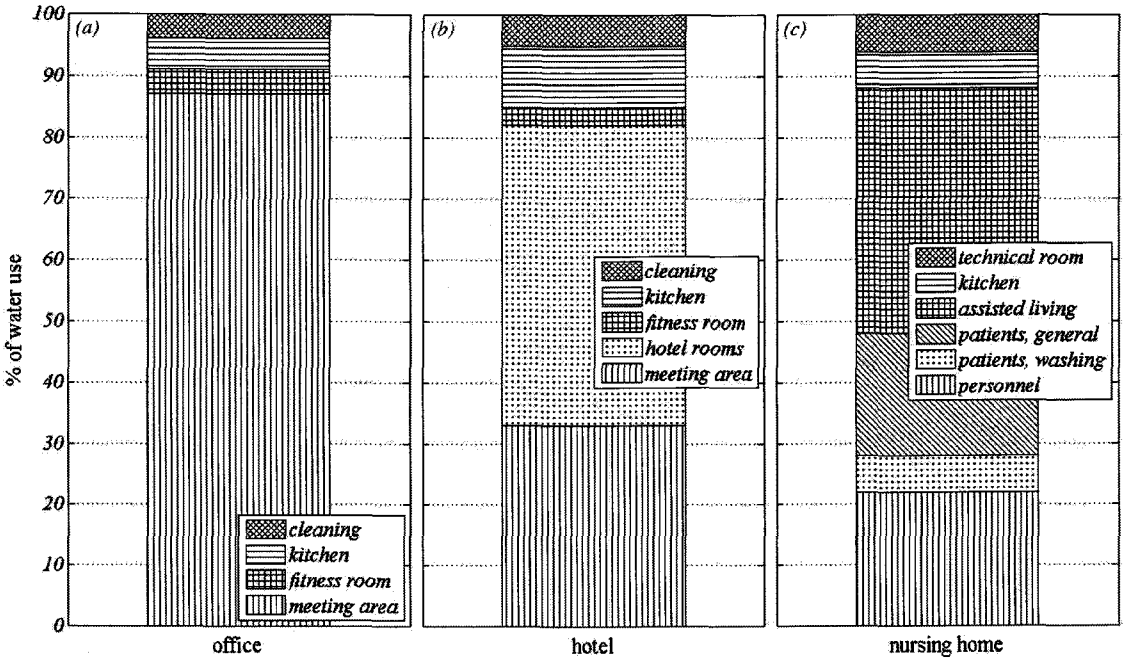


Fig. 8. Percentage of total water use per functional room for (a) an office building, (b) a hotel, and (c) a nursing home

and residents and by the showering of residents (patients and people in assisted-living residences).

The end-use model uses less stochastic information for the nonresidential water demand than for the residential demand. For example, the number of water-using appliances is a fixed number, and the users are simulated as a group of users rather than as individual users. The diurnal pattern for the probability of a water-use events is specific per functional room, and is valid for all users alike. This approach, using a specific diurnal pattern, appears to work well. The parameters for the start and end times for user presence and greater probabilities of water use were collected from only a few surveys, and are not yet generalized for all functional rooms. Nevertheless, the end-use model works well. The assumptions on the frequency of use, duration, and flow of water-use events and the probability of when such an event is most likely to occur seem to be estimated well. The fact that the end-use model is based on prior knowledge of water demand, rather than on water demand measurements, and the fact that the figures show that the model leads to good simulation results, is very promising for other types of nonresidential buildings with many people using water that share some functional rooms with the building types described in this paper, e.g., penitentiaries.

The basics of the end-use model for nonresidential demand were completely developed. However, compared to the end-use model for residential water demand, the end-use model for nonresidential water demand was not fully expanded. The information on nonresidential water use is quite specific for the particular offices, hotels, and nursing homes that were studied in this project because little generic information could be found in literature, and it was difficult to derive generics from the limited project-specific surveys. Also, the validation of the nonresidential water-demand model was only done for the limited number of buildings that also filled out the surveys. More validation is recommended for other types of nonresidential buildings for which no completed surveys were available.

The end-use model for nonresidential water demand can be used to design the domestic hot and cold water system of nonresidential buildings (Pieterse-Quirijns et al. 2010) or the drinking-water distribution system to which nonresidential buildings are connected. Also, scenario studies can be done to study the effect of changes in behavior or water appliances on total and maximum water demand. Water distribution network models can be populated with realistic demands of all types of residential and nonresidential buildings; this will improve the study of water quality in the distribution network (Blokker 2010). Depending on the purpose, different timescales are important. In the design stage, water demand per second should be considered (Pieterse-Quirijns et al. 2010); in water-quality studies, a timescale of 5 or 15 min may be sufficient (Blokker et al. 2011).

Conclusions

The end-use model SIMDEUM, which was developed to predict residential water demand, was extended to predict nonresidential water demands. The model could easily be extended because information on the water-using appliances and water users is its input. The model uses functional rooms that can be used in different types of nonresidential buildings, not just the office, hotel, and nursing home presented.

The water use in office buildings is primarily determined by the flushing of toilets. It is therefore important to have reliable information on toilet use. The water use in hotels (both total use and diurnal pattern) is primarily determined by the use in the hotel rooms. It is therefore important to have reliable information on the number and occupancy of the rooms and the water-using appliances in the rooms. The water use in the presented nursing home is not determined by a single dominant functional room.

First results show that the simulation results are in good agreement with measured water-demand patterns. This is true for a range of timescales (from 5 min to 1 h) and a spatial scale of a single office, hotel, and nursing home. The assumptions on the frequency

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of use, duration, and flow of water-use events, as well as the probability of when such an event is most likely to occur, thus seem to be estimated well. The results are promising; the model may also be applied for other types of nonresidential buildings.

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